Memtrix - How to Play

Getting Started

1. **Launch the Game**:
    - Ensure you have Python and Pygame installed.
    - Run the game by executing main.py (e.g., python main.py).
    - The game is also compatible with WebAssembly environments via Pyodide/Emscripten.
2. **Main Menu**:
    - Upon starting, you'll see the **Difficulty Selection** screen with three options: **Easy**, **Medium**, and **Hard**.
    - Click one of the difficulty buttons to begin the game.

Game Mechanics

Difficulty Levels

Memtrix offers three difficulty levels, each affecting the grid size, timer duration, and maximum level:

- **Easy**:
    - Initial grid: 2x2 (4 tiles).
    - Timer: 90 seconds (60s × 1.5 multiplier).
    - Maximum grid: 4x4 (16 tiles).
    - Slower tile display for easier memorization.
- **Medium** (default):
    - Initial grid: 4x4 (16 tiles).
    - Timer: 60 seconds.
    - Maximum grid: 8x8 (64 tiles).
- **Hard**:
    - Initial grid: 5x5 (25 tiles).
    - Timer: 48 seconds (60s × 0.8 multiplier).
    - Maximum grid: 10x10 (100 tiles).
    - Faster pace and more complex patterns.

Gameplay Flow

1. **Difficulty Selection**:

- From the main menu, click **Easy**, **Medium**, or **Hard** to start the game with the chosen difficulty.
- The game transitions to the **Playing** screen.

2. **Memorizing the Pattern**:
   - A grid of tiles (e.g., 4x4 for Medium) is displayed.
   - A sequence of tiles lights up in **green** one at a time, indicating the pattern you must memorize.
   - The number of tiles in the pattern equals the current level (e.g., 4 tiles for a 4x4 grid).
   - A prompt ("Memorize this pattern") appears at the bottom of the screen.
   - The display duration for each tile is approximately 0.8 seconds (longer for Easy difficulty), followed by a brief pause.
3. **Replicating the Pattern**:
   - After the pattern is shown, click the tiles in the same order they were highlighted.
   - Correct clicks highlight tiles in **green**; incorrect clicks highlight tiles in **red**.
   - You must complete the sequence within the timer (visible as a green bar at the bottom of the screen).
4. **Scoring and Progression**:
   - Successfully replicating the pattern earns **10 points** and resets the timer.
   - The game advances to the next level, increasing the grid size (e.g., from 4x4 to 5x5) and pattern length.
   - The tile size shrinks slightly with each level, making the grid denser and harder to navigate.
   - If you reach the maximum grid size for your difficulty (e.g., 8x8 for Medium), the game continues with longer patterns but no further grid expansion.
5. **Game Over**:
   - The game ends if:
     - You click a tile out of sequence (incorrect pattern).
     - The timer runs out (timer bar depletes).
   - The **Game Over** screen displays your score and the high score for the selected difficulty.
   - From here, you can:
     - Click **Play Again** to restart with the same difficulty.

        - Click **Leaderboard** to view high scores across all difficulties.

6. **Pausing the Game**:
    - Press the **Escape** key during gameplay to enter the **Pause** screen.
    - From the Pause screen, click **Back to Menu** to return to the difficulty selection screen, or press **Escape** again to resume.
7. **Leaderboard**:
    - The **Leaderboard** screen shows high scores for each difficulty (e.g., Easy: 1, Medium: 2, Hard: 3).
    - Click **Play Again** to start a new game.

Controls

- **Mouse Click**: Select difficulty, click tiles to replicate patterns, or interact with buttons (e.g., Play Again, Leaderboard).
- **Escape Key**: Pause or resume the game.

User Interface

- **Visual Feedback**:
    - **Tiles**: Highlight in green during pattern display and correct input, red for incorrect input.
    - **Timer Bar**: A green bar at the bottom shows remaining time, shrinking as the timer counts down.
    - **Score and Level**: Displayed at the top (e.g., "Score: 20", "Level: 4", "Best: 30").
    - **Animations**: Tiles scale and fade when highlighted, and the background features a scrolling neon gradient.
- **Screens**:
    - **Difficulty Selection**: Neon-green title ("Memtrix") with buttons for Easy, Medium, and Hard.
    - **Playing**: Grid with animated tiles, timer bar, and score/level display.
    - **Game Over**: Shows score, high score, and options to restart or view the leaderboard.
    - **Leaderboard**: Lists high scores by difficulty.
    - **Pause**: Displays "Paused" with a Back to Menu button.

Scoring and High Scores

- **Scoring**: Each correct pattern earns 10 points.
- **High Scores**: Tracked per difficulty (Easy, Medium, Hard) and saved to highscores.json.

- **Example**: If you score 20 on Medium, it updates the high score if higher than the previous value (e.g., from 2 to 20).

Tips for Success

- **Focus on the Pattern**: Watch the sequence carefully, as tiles light up one at a time.
- **Use Visual Cues**: The green highlights and animations help track the order.
- **Manage Time**: The timer resets on correct patterns, so prioritize accuracy over speed.
- **Start with Easy**: If you're new, try Easy mode to get familiar with the mechanics before tackling Hard.
- **Pause Strategically**: Use the Escape key to pause if you need a break during gameplay.

System Requirements

- **Desktop**:
    - Python 3.x
    - Pygame library (pip install pygame)
    - Operating System: Windows, macOS, or Linux
- **Web**:
    - Compatible with Pyodide/Emscripten for browser-based play.
    - No additional installation required if hosted on a compatible platform.

Known Limitations

- **No Sound**: The game lacks audio feedback (e.g., sound effects for clicks or music).
- **Accessibility**: Limited support for colorblind players (relies on green/red colors) or adjustable font sizes.
- **No Tutorial**: New players may need a few tries to understand the mechanics.

Troubleshooting

- **Game Won't Start**: Ensure Pygame is installed (pip install pygame) and run main.py with Python 3.x.
- **High Scores Not Saving**: Check that highscores.json is writable in the game directory.
- **Slow Performance**: On low-end systems, the animated background may cause lag. Try reducing the window size in settings.py (e.g., WIDTH = 400, HEIGHT = 400).